

Anshul Shah

412-585-5944 | anshul7@uw.edu | linkedin.com/in/anshul-shah1/ | github.com/anshll | anshll.github.io/

EDUCATION

University of Washington

Seattle, WA

Bachelor of Science in Computer Science | GPA: 3.91

Sept 2023 – June 2027

Coursework: Distributed Systems, Machine Learning, Reinforcement Learning (grad), Operating Systems, Algorithms, Probability & Statistics, Data Structures & Parallelism, Linear Algebra, Differential Equations, Wireless Communication

TECHNICAL SKILLS

Language/Framework: Java, Python, JavaScript/TypeScript, HTML/CSS, C/C++, Node.js, React, Svelte

Tools/Technologies: WebAssembly, Jupyter Notebooks, Web Sockets, REST APIs, Git, Docker, Unit Testing

Libraries: NumPy, tensorflow, Torch, transformers, Matplotlib, scikit-learn, PyAudio, hashlib, nltk, Express, Flask

EXPERIENCE

Baker Lab

April 2026 – Present

Undergraduate Machine Learning Researcher

Seattle, WA

- At Nobel prize winning lab fusing protein generative models with LLM architectures to add reasoning capabilities

Amazon (AWS)

May 2025 – August 2025

Software Development Engineer Intern

Seattle, WA

- Built a serverless, AI-powered system to automate onboarding new AWS resource types (RTs) to AWS Config — a **day-long manual process now 2 minutes (150× speedup)**. Invoked via ApiGateway endpoint, outputs in Dynamo, S3
- Grounded Bedrock LLMs in documentation to author XML specs (each with dynamic tags/attributes and complex inline JSLT transformations) for mapping describe APIs into AWS Config configuration items.
- Created feedback loop with existing tests to let LLM self-correct until specs passed. On **30 RTs, 100% test accuracy**.

Dartmouth Health

June 2024 – Sept 2024

Machine Learning Intern

Hanover, NH

- Assessed literature, designed, and trained a BERT based model for Named Entity Recognition (NER) of phenotypes in clinical notes with PyTorch — evaluated by a geneticist to automate **80% of manual annotation work**
- Combined techniques from **state-of-art models PhenoBCBERT & PhenoID**: IOB labeling for more reliable annotations, window+striding for larger context, and 2nd finetuned BERT to normalize to phenotype IDs in ontology.
- Deployed Flask microservice/API to serve model, with React **user-friendly annotation interface** for error correction

Dubhacks NEXT

June 2024 – Present

Co-Managing Director

Seattle, WA

- Organize UW's student-run startup incubator; our startups have **raised \$2M cumulatively**, multiple at YCombinator
- Lead sponsorship outreach, secured **\$35K from Amazon, Hashicorp, PackVC, Madrona Ventures** over 2 years

UW Sensor Systems Lab

Nov 2024 – May 2025

Paid Undergraduate Research Assistant

Seattle, WA

- Predicted wireless signal strength of a lunar rover for a **NASA and Astrobotic funded project** w/ Prof. Josh Smith
- Created new Monte-Carlo-like metric and tuned its hyperparameters** to summarize a terrain's wireless properties
- Increased path loss **accuracy by $\approx 20\text{dB}$** , translating electromagnetic properties of lunar soil in literature to simulation

PROJECTS

Distributed Key-Value Store | Java

March 2026

- Built a **linearizable KV store sharded** across Multi-Paxos replica groups with a dedicated controller for rebalancing
- Implemented **fault-tolerant** Paxos leader election, log slot proposals, merging, and garbage collection plus shard rebalancing logic; all messages **resilient** against simulated network partitions, drops, delays, and duplicates

Renaissance Research | Python, Beautiful Soup, AWS Bedrock, Gemini, Pydantic, Next.js

April 2025

- Created software that scrapes old/niche research papers, and uses **multiple agents** through Bedrock Batch Processing and Gemini to evaluate work on potential novel modern day applications, with a Next.js interface for viewing results
- Won **grand prize at Caltech Hackathon (2x Title Track Winner, \$25,000 OpenAI credits, NVIDIA RTX 5070 GPUs)**

Roulette | Javascript, Webassembly, Rust, HTML, CSS

Jan 2024 – June 2024

- Built service for websites to **monetize by lending visitors' compute** for large parallel Monte Carlo Simulations
- Sped up simulation code **20x faster than JavaScript** with WebWorkers running WebAssembly compiled from Rust